IE 643: Deep Learning - Theory and Practice **July-Dec 2026**

Project Report: Multivariate Weather Forecasting

Team Name: The Epoch-alypse Survivors *Team Members: 24B1013 , 24B0995*

Abstract

This project report describes our work on multivariate weather forecasting with calibrated uncertainty, with a specific focus on extreme (hot) events. Given 72 past six-hourly steps of five ERA5 variables (temperature, dew point, wind speed, cloud cover and precipitation), our Quantile CNN-LSTM predicts the 5%, 50% and 95% quantiles of the temperature anomaly for the next 12 steps, so that a 90% prediction interval comes directly from the network. Extreme events are defined by a peaks-over-threshold criterion (95th percentile of the training-period anomaly). We pretrain the network on 72 ERA5 grid points and adapt it to 9 held-out grid points with parameter-efficient fine-tuning, LoRA and DoRA, and compare them against zero-shot transfer, training from scratch and full fine-tuning. We additionally apply split-conformal calibration of the intervals. In our current evaluation on the 2023 test year, LoRA and DoRA, which train only 10.2% and 12.2% of the 46,460 parameters, reach the lowest extreme-event RMSE (0.127 and 0.128, against 0.131–0.135 for the other models) and the highest extreme-event interval coverage (approximately 0.72), while normal-event accuracy is essentially unchanged (RMSE 0.082–0.084) and normal-event coverage is 0.92–0.93. Coverage on extreme events nevertheless remains well below the nominal 0.90, which we report as the main limitation. These findings are a combination of 3 seeds for better claims.

1 Introduction

Weather forecasts are used for decisions in agriculture, energy, transport and disaster management, and the cost of a forecast error is not uniform: it is the highest during extreme events such as heat waves, when a point forecast with no indication of confidence can be misleading. Two properties are therefore needed together: an accurate median forecast, and an *uncertainty estimate that can be trusted* exactly when conditions are unusual. This is difficult because extreme events are rare, so a model trained to minimize the average error sees few of them, and a prediction interval that is well calibrated on average is many times too small for the extremes.

Deep learning is an excellent fit for this task. Multivariate weather series are non-linear and non-stationary, with interactions between temperature, humidity, wind and cloud that linear statistical models (e.g. VAR, ARIMA) capture poorly. Convolutional layers extract local temporal patterns, recurrent LSTM layers carry longer-term state, and a quantile (pinball) loss lets the same network output a prediction interval directly, without a Gaussian assumption on the errors. Finally, because weather at neighboring places is strongly related, a model pretrained on many grid points can be reused for a new location by training only a small number of extra parameters (LoRA/DoRA), which is attractive when the target location has little data or when many locations must be served.

In this project we (i) build a forecasting pipeline on ERA5 reanalysis data, (ii) define extreme events with a statistically motivated peaks-over-threshold criterion, (iii) train a Quantile CNN-LSTM that outputs a 90% prediction interval, (iv) adapt a pretrained model to new locations with LoRA and DoRA and compare it with full fine-tuning and training from scratch, and (v) evaluate accuracy and calibration separately on normal and extreme events.

We provide the project workflow in Section 2. A survey of existing literature is given in Section 3. Our proposal for the project is described in Section 4, and the data in Section 5. We give details on experiments in Section 6 and results in Section 7. A description of future work is given in Section 8. We conclude with a short summary and pointers to forthcoming work in Section 9.

2 Project Workflow

The project progressed through the following steps (Figure 1 summarizes them).

- **Problem statement and literature review.** We defined the task as multi-step multivariate temperature forecasting with calibrated 90% intervals, evaluated separately on normal and extreme events, and surveyed quantile deep learning, CNN-LSTM weather forecasting and ERA5 Dataset(Section 3).
- **Data selection and design (before the prep-presentation).** We chose ERA5 reanalysis over single-station and raw network data, selected the variables, designed the preprocessing, chose the peaks-over-threshold definition of extremes, chose the Quantile CNN-LSTM architecture and the metrics (RMSE, MAE, PICP, MPIW), and planned how to represent all of this.
- **Prep-presentation review.** We presented the design above. After the review, the instructors/TA asked us to use a pretrained quantile CNN-LSTM and to apply LoRA and DoRA to it.
- **First implementation.** We could not find a pretrained quantile CNN-LSTM model which we could use. Thus, we implemented the whole pipeline in one script: ERA5 download via the CDS API, preprocessing the data, a single-stage Quantile CNN-LSTM, evaluation, and a web interface.
- **Diagnosis.** Reading the first results and code critically, we found data leakage, poor deseasonalisation, an inconsistent extreme-event definition and an untrained model being used in the web interface (Section 4.2).
- **Re-engineering.** We rebuilt the pipeline: train-period-only statistics, weather anomalies, a pretrain-then-adapt method with held-out target locations, hand-written LoRA/DoRA adapters, baselines (zero-shot, scratch, full fine-tuning), split-conformal calibration, early stopping and a real-model web interface.
- **Iteration.** The first re-engineered run showed a weak pretrained model (validation loss rising after epoch 1). We lowered the pretraining learning rate and increased patience, and re-ran.
- **Evaluation and reporting.** Metrics were computed on 2023 for the 9 target locations, split into normal and extreme subsets.

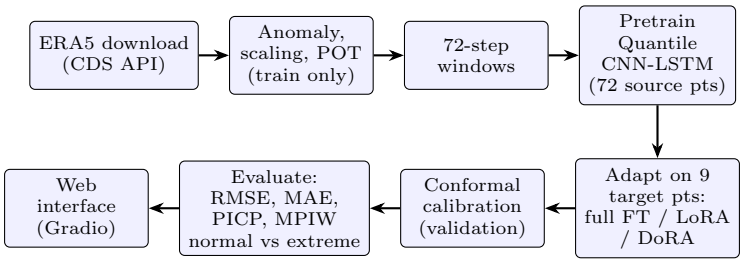


Figure 1: Project workflow from raw ERA5 data to calibrated forecasts and the web interface.

3 Literature Survey

This section lists the works that were most useful to us. Each paper's main contribution, and how it relates to our project, is summarised below.

Quantile deep learning for multi-step forecasting. Our project is closely related to the work of Cheung et al. [2], who add a quantile (pin-ball) loss to LSTM-family models (bidirectional, encoder-decoder and convolutional LSTMs) for multi-step prediction, outputting the quantiles 0.05, 0.25, 0.5, 0.75 and 0.95 and using the median as the point forecast. They evaluate on cryptocurrency series (Bitcoin, Ethereum) and on benchmark series (Sunspot, Mackey-Glass, Lorenz), and report that quantile versions are about as accurate as standard versions (differences in RMSE are small) while additionally giving interval information. Their evaluation reports RMSE only: the coverage of the predicted quantiles is not measured, and no separate evaluation on extreme events is made. We adopt their quantile-output idea and add PICP/MPIW calibration metrics and a separate extreme-event evaluation.

CNN-LSTM for weather and temperature. Kadel and Kalla [1] proposed a CNN-LSTM framework for multivariate temperature forecasting and extreme weather event analysis, which is the closest application to ours. Shen et al. [3] predict temperature indicators in eastern China with a multi-scale CNN-LSTM-attention network on hourly records (2001–2020) with a 30-day input window, and report point-forecast error (MSE/RMSE) only; no uncertainty or extreme-event evaluation is reported. Jha and Rathore [4] describe a hybrid CNN-LSTM framework for climate prediction from environmental data. These works led to our choice of a CNN model with an LSTM, because of its high accuracy and wide usage in past research papers.

Probabilistic forecasting with the pinball loss. Wang et al. [6] use a pinball-loss-guided LSTM for probabilistic individual load forecasting, showing that quantile outputs give a distribution-free uncertainty estimate for sequence models. The pinball loss goes back to quantile regression [12]. We use the same loss for temperature anomalies.

Transformers for time series. Zeng et al. [5] show that simple linear models can outperform Transformer-based forecasters on long-horizon benchmarks, because self-attention is permutation-invariant and loses temporal order. This supported our choice of a convolutional-recurrent model over a Transformer for this problem.

Parameter-efficient fine-tuning. LoRA [8] freezes pretrained weights and learns a low-rank update $\Delta W = BA$ to selected weight matrices, so only a small fraction of parameters is trained. DoRA [9] decomposes each weight into a magnitude and a direction and applies the low-rank update to the direction, with a trainable magnitude vector. Both were introduced for large language models. We thus use these techniques to adapt our model to unseen locations.

Conformal prediction. Conformalized quantile regression [10] turns the quantile outputs of any model into intervals with a finite-sample marginal coverage guarantee, by widening them by a quantile of nonconformity scores computed on held-out calibration data. Adaptive conformal inference [11] extends this to distribution shift, which is relevant for us because our test year is warmer than the training years (Section 5).

Extreme value theory and data. The peaks-over-threshold (POT) method [13] classifies extreme values. ERA5 [7] is the reanalysis we use as data. The POT illustration [14] and the LSTM diagram [15] were used in our prep-presentation.

4 Proposed Approach or Approaches

Our approach differs from the works above in four respects: (1) it evaluates calibration (PICP, MPIW) and accuracy separately on normal and extreme events; (2) it uses a pretrain-then-adapt protocol with held-out locations, and compares LoRA/DoRA with zero-shot, scratch and full fine-tuning; (3) it applies conformal calibration to the quantile intervals; and (4) every statistic (climatology, scaling, threshold, calibration margin) is computed on data that precedes the data it is applied to.

4.1 Work done before prep-presentation review

Before the review, we designed the project (the content of our prep-presentation).

Task. Forecast future weather variables from multivariate time series that contain normal and extreme events, provide uncertainty estimates, identify extreme events with a statistically justified criterion, and check whether the predicted intervals reflect the true uncertainty, especially under extreme conditions.

Data choice. We compared (a) single-station datasets such as the Jena climate data (no spatial structure, few extreme events), (b) raw station networks such as NOAA ISD/GSOD (broken sensors, missing values, uneven timestamps, inconsistent density) and (c) ERA5 reanalysis (benchmark, hourly, 31 km resolution, assimilates observations with numerical weather models, easy to slice by region). We chose ERA5 due to its credibility.

Preprocessing plan. Select the region; use temperature, dew point, cloud cover, precipitation, wind speed and wind direction; convert wind speed and direction into continuous u/v components; fill gaps by interpolation; min–max scale each feature to $[0, 1]$; and build sliding-window tensors (a window of 72 steps to forecast a future window).

Extreme events. Define extremes with a threshold at the 95th percentile. We considered full Extreme Value Theory, but a fitted tail model is not required when we want to output an interval, and the 95th-percentile threshold is the first stage of the peaks-over-threshold method [13, 14] so we settled on that.

Architecture choice: Quantile CNN-LSTM. We compared (i) statistical baselines (VAR, ARIMA), which assume linearity, scale badly across weather grids, need manual lag engineering and give no native uncertainty bounds; (ii) Transformers (Informer, PatchTST), which capture global patterns but lose temporal order through attention, mismatch continuous series and accumulate error in multi-step roll-outs [5]; and (iii) the Quantile CNN-LSTM, which tracks sequences with memory, maps directly to multiple horizons (avoiding compounding error), gives uncertainty natively through the pinball loss (no Gaussian assumption), and can widen intervals in volatile periods [2, 6].

Components and their roles.

- *1-D convolution (+ReLU, max-pooling):* extracts local temporal patterns across the variables and halves the sequence length before the recurrent layers.
- *LSTM:* a long-term memory state with forget, input and output gates carries the momentum of past weather; the forget gate drops irrelevant old patterns, the input gate admits informative new observations, and the output gate combines memory with the current input.
- *Multi-quantile output layer:* a linear layer produces, for every forecast step, one value per quantile, so the three quantiles form a prediction interval and the median is the point forecast.
- *Pinball loss:* the asymmetric loss that makes each output converge to its conditional quantile.

Metrics. RMSE and MAE for deterministic accuracy, PICP for interval reliability and MPIW for interval sharpness.

Dashboard plan. A user interface for regional data, with shaded intervals and highlighted extremes (planned in Streamlit/Plotly; implemented later in Gradio).

4.2 Work done after prep-presentation review

After the review, the guidance was to use a pretrained Quantile CNN-LSTM and adapt it with LoRA and DoRA. We first implemented a single-stage version of the pipeline (training from scratch for a few epochs, which we call the *initial baseline*), and then re-engineered it.

4.2.1 Initial baseline and the problems found

The initial baseline used four features (temperature, wind speed, cloud cover, precipitation), subtracted a 30-step centered rolling mean as a "seasonal" component, fitted the min–max scaler, the 95th percentile and a generalised Pareto fit (99% quantile of the exceedances) on the entire series, and trained one model for 5 epochs at learning rate 5×10^{-3}. Its web interface created an untrained model and fed it random noise. We identified the following problems:

- **Data leakage:** scaler, percentile and tail threshold used the test period.
- **Weak deseasonalisation:** a 30-step window on six-hourly data spans about 7.5 days, not a seasonal cycle.
- **Inconsistent extreme definition:** a 99% GPD quantile on top of the 95th percentile, different from the 95th-percentile rule in our slides, and leaving too few extreme targets.
- **No pretraining or adaptation,** We did not find any pretrained model of the same architecture that we could use.

4.2.2 Final pipeline

Climatological anomaly. For each location and variable we compute the mean of the training years for every (day-of-year, hour-of-day) pair, smooth it over ± 7 days with circular padding, and subtract it from all years, giving an anomaly series a_t. Anomalies are min–max scaled with the training-period minimum and maximum.

Extreme events (peaks over threshold). Let u be the 95th percentile of the training-period temperature anomaly at a location. A target step is *extreme* if $a_t > u$. We decided to drop the generalized Pareto distribution fit to the training extremes $a_t - u$ as a tail diagnostic.

Windows. Input $X \in \mathbb{R}^{72\times 5}$ (18 days), target $y \in \mathbb{R}^{12}$: the scaled temperature anomaly for the next 12 steps (3 days).

Backbone. The model is

$$h = \mathrm{MaxPool}_2\big(\mathrm{ReLU}(\mathrm{Conv1D}_{5\to 64,k=3}(X))\big), \tag{1}$$

$$z = \mathrm{LSTM}_{64\to 50,\,2\text{ layers}}(h)_{\text{last step}}, \tag{2}$$

$$\hat{q} = \mathrm{Linear}_{50\to 12\times 3}(z) \ \in \mathbb{R}^{12\times 3}, \tag{3}$$

with 46,460 trainable parameters (convolution 1,024, LSTM 43,600, output layer 1,836). The outputs are sorted along the quantile axis at inference so that $\hat{q}_{0.05} \le \hat{q}_{0.5} \le \hat{q}_{0.95}$.

Pinball loss. For quantile τ and error $e = y - \hat{q}_\tau$, $L_\tau = \max(\tau e, (\tau - 1)e)$, and the training loss is the average over the three quantiles and 12 steps,

$$\mathcal{L} = \frac{1}{3} \sum_{\tau \in \{0.05, 0.5, 0.95\}} \frac{1}{12} \sum_{k=1}^{12} w_k \, L_\tau\big(y_k, \hat{q}_{\tau,k}\big), \tag{4}$$

where the weight $w_k = 1$ for normal targets and $w_k = 3$ for extreme targets during adaptation (tail-weighted loss).

Pretrain then adapt. The 81 grid points are split into 72 *source* and 9 *target* points (fixed random seed). The base model is pretrained on the source points and then adapted on the target points. Five variants are compared:

- **Zero-shot:** the pretrained model with no target training.
- **Scratch:** a randomly initialised model trained on the target data only (a control for the value of pretraining).
- **Full fine-tuning:** the pretrained model, all 46,460 parameters trained.
- **LoRA:** pretrained weights frozen; for each adapted weight $W \in \mathbb{R}^{d_{out} \times d_{in}}$ we learn $B \in \mathbb{R}^{d_{out} \times r}$ (initialised to zero) and $A \in \mathbb{R}^{r \times d_{in}}$, and use

$$W' = W + \frac{\alpha}{r} BA. \tag{5}$$

- **DoRA:** the same low-rank update, but with the row-wise decomposition into magnitude $m \in \mathbb{R}^{d_{out}}$ (initialised to the row norms of W) and direction,

$$W' = m \odot \frac{W + \frac{\alpha}{r} BA}{\|W + \frac{\alpha}{r} BA\|_{\text{row}}}. \tag{6}$$

The adapters are applied to the convolution kernel, both LSTM layers (input-to-hidden and hidden-to-hidden matrices) and the output layer, with rank $r = 4$ and $\alpha = 8$; the output-layer bias is also trained. We use PyTorch's functional parameter replacement because `nn.LSTM` reads its weights from internal tensors, so adapters cannot be attached as separate sub-layers. This gives 4,752 trainable parameters for LoRA (10.2%) and 5,652 for DoRA (12.2%). At initialisation both adapted models are exactly equal to the pretrained model (verified numerically).

Split-conformal calibration. On the validation set of the target locations we compute nonconformity scores $s = \max(\hat{q}_{0.05} - y,\ y - \hat{q}_{0.95})$ for every element, take their $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile $\hat{s}$ with $1 - \alpha = 0.90$, and widen every test interval to $[\hat{q}_{0.05} - \hat{s},\ \hat{q}_{0.95} + \hat{s}]$ (conformalized quantile regression [10]). This is reported as "+conformal".

Metrics. For truth y, median forecast $\hat{m}$ and interval $[\ell, u]$ over N elements:

$$\text{RMSE} = \sqrt{\tfrac{1}{N} \textstyle\sum (y - \hat{m})^2}, \quad \text{MAE} = \tfrac{1}{N} \textstyle\sum |y - \hat{m}|, \tag{7}$$

$$\text{PICP} = \tfrac{1}{N} \textstyle\sum \mathbb{I}(\ell \le y \le u), \quad \text{MPIW} = \tfrac{1}{N} \textstyle\sum (u - \ell). \tag{8}$$

The nominal PICP of a 5%–95% interval is 0.90. All metrics are computed separately for normal and extreme target steps.

Web interface. A Gradio page loads the adapted DoRA model and shows, for a chosen test window, the history, the true future, the median forecast, the (conformally widened) 90% interval and the extreme steps. A checkbox restricts the selection to windows that contain extreme events.

5 Data set Details

Source. We use ERA5 hourly data on single levels [7], produced by ECMWF and distributed by the Copernicus Climate Data Store (CDS) at `https://cds.climate.copernicus.eu` (dataset: `reanalysis-era5-single-levels`). Data were requested programmatically with the `cdsapi` Python client, one year per request, and cached locally as NetCDF files.

Nature and size. The data are numerical gridded time series. Our request covers the region 18°–20°N, 72°–74°E (a $2° \times 2°$ box, 81 grid points on the 0.25° ERA5 grid), at 00, 06, 12 and 18 UTC, for the years 2015–2023. Each location has 13,148 time steps, for a total of 10,64,988 rows.

Attributes. The variables requested are 2 m temperature, 2 m dew-point temperature, 10 m u and v wind components, total cloud cover and total precipitation. After preprocessing there are five input channels: Temperature, Dewpoint, Wind_Speed ($= \sqrt{u^2 + v^2}$), Cloud_Cover and Precipitation. The prediction target is the temperature anomaly. (The prep-presentation also listed wind direction; we use the u/v components through wind speed instead.)

Preprocessing.

1. Extract the NetCDF files (new CDS downloads are zip archives), merge variables on (location, time), compute wind speed, build a location identifier from latitude and longitude, and fill any remaining missing value forward and backward within each location.
2. Per location, compute the climatology on the training years only and subtract it to obtain anomalies (Section 4.2).
3. Min–max scale each anomaly channel with training-period statistics.
4. Flag extreme target steps with the 95th-percentile threshold of the training-period temperature anomaly.
5. Build sliding windows: 72 input steps and 12 target steps.

Splits. The split is chronological, by the time of the last target step of each window: train 2015–2020, validation 2021-2022, test 2023. The 81 locations are split spatially into 72 source and 9 target points. Window counts are given in Table 1.

Table 1: Number of windows and share of extreme target steps in each split.

	Train (2015–20)		Val (2021–22)		Test (2023)	
Domain	windows	extreme	windows	extreme	windows	extreme
Source (72 pts)	625,320	5.04%	210,240	7.29%	105,120	9.18%
Target (9 pts)	78,165	5.04%	26,280	7.47%	13,140	8.62%

The training is done using the training windows and the validation windows of the source. The adaptations factor in the target training windows and validation windows. The final testing is done on the test windows of the target locations. Hence, the source locations of the test year are not used anywhere. The share of extreme steps rises from about 5% in training to about 8.6% in 2023 because the threshold is fixed from the training years and the test year is warmer. This distribution shift matters for calibration

(Section 7). The fitted generalised Pareto shape parameter over locations has mean $\xi = -0.115$ (range -0.309 to 0.093), indicating a mostly bounded tail.

6 Experiments

Code. Our code is available at https://github.com/JiyaGada/IE-643-Course-Project. It contains the ERA5 download script, the training and evaluation pipeline, the LoRA/DoRA implementation and the web interface. Packages used: PyTorch, `cdsapi`, xarray, NumPy, pandas, SciPy and Gradio.

Hardware. The machine used had: Intel Core i9-13900H (2.60 GHz), 16 GB RAM, Intel Iris Xe Graphics, Windows 11 Home (25H2), x86-64 architecture. All experiments were run on a Windows personal computer using a Python virtual environment.

Training algorithm. Algorithm 1 summarises the procedure. Optimisation uses Adam throughout, mini-batch size 256, and early stopping on a validation pinball loss computed on an evenly spaced subset of at most 20,000 validation windows; the best-validation weights are restored at the end.

Algorithm 1 Pretrain, adapt and calibrate

1: Compute climatology, scaling and POT threshold on the training years of each location
2: Build windows; split chronologically and into source/target locations
3: Pretrain f_θ on source-train with Adam (lr 5×10^{-4}, $\leq$ 15 epochs, patience 6); keep best validation weights
4: **for** each method in {Scratch, Full FT, LoRA, DoRA} **do**
5: Initialise from f_θ (Scratch: random); freeze weights except adapters (LoRA/DoRA)
6: Train on target-train with the tail-weighted pinball loss (Eq. 4), ≤ 20 epochs, patience 4
7: Compute the conformal margin $\hat{s}$ on target-validation
8: Evaluate on target-test: normal and extreme subsets, raw and +conformal

Optimisation settings. Pretraining: learning rate 5×10^{-4}. Scratch: 2×10^{-3}. Full fine-tuning: 5×10^{-4}. LoRA and DoRA: 2×10^{-3} (adapter parameters only), rank $r = 4$, $\alpha = 8$, tail weight 3. The random seed is fixed for each run; the 9 target locations are the same in all runs.

Experiments performed.

- **E0: initial single-stage baseline** (before re-engineering): four features, rolling-mean deseasonalisation, statistics fitted on the full series, 5 epochs at learning rate 5×10^{-3}, batch size 128, no pretraining. Outcome: normal events PICP 0.81, MPIW 0.16, RMSE 0.07, MAE 0.05; extreme events PICP 0.00, MPIW 0.18, RMSE 0.23, MAE 0.23. These numbers are not comparable with later ones because the scaling, the anomaly definition and the extreme definition differ and the pipeline leaked test information; we report them as a negative result that led to the redesign.
- **E1: first re-engineered pipeline** (pretraining learning rate 2×10^{-3}, patience 4, up to 12 epochs). The validation loss was best at epoch 1 and then rose while the training loss kept falling (Figure 2), so the saved pretrained model was effectively a one-epoch model. Results are in Table 2.
- **E2: current pipeline** (pretraining learning rate 5×10^{-4}, patience 6, up to 15 epochs), three seeds. Results are in Tables 3 and 4.
- **E3: extreme event definition tweaked** Instead of considering only the hot events, we also considered the cold ones, i.e. defined extreme events to be values above the 95 percentile or below the 5th percentile. This gave us 10 percent extremes in the training windows, and 13-14 percent extremes in the validation and testing windows.

7 Results

All numbers below are on the 9 target locations in the 2023 test year, on the scaled temperature anomaly (so RMSE, MAE and MPIW have no physical units). The nominal PICP is 0.90. **All numbers in Tables 2–4 come from 3 random seeds (0,1,2), so differences of a few thousandths should not be interpreted as a ranking.**

7.1 Effect of the pretraining quality

In E1 (Table 2) the pretrained model was weak, and training from scratch on the target locations gave the best normal-event accuracy (RMSE 0.097 against 0.113 zero-shot). Figure 2 shows why: validation loss rose from epoch 2 while training loss kept falling, which we attribute to a too large learning rate and to a climatology fitted on only two years. In E1 the zero-shot model had the lowest extreme RMSE (0.132), and all adapted models were worse than zero-shot on extremes.

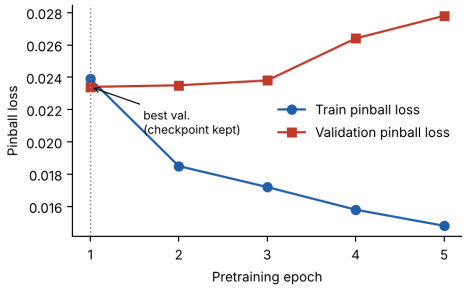


Figure 2: E1 pretraining curve (learning rate 2×10^{-3}): the validation pinball loss is lowest at epoch 1 and rises afterwards, while the training loss falls.

Table 2: E1 (weak pretrained base): accuracy and calibration, single seed.

		Normal				Extreme			
Model	Train. params	RMSE	MAE	PICP	MPIW	RMSE	MAE	PICP	MPIW
Zero-shot	0	0.113	0.084	0.880	0.322	0.132	0.110	0.732	0.321
Scratch	46,460	0.097	0.072	0.925	0.350	0.138	0.117	0.820	0.356
Full FT	46,460	0.106	0.079	0.890	0.326	0.153	0.131	0.710	0.331
LoRA (r=4)	4,752	0.108	0.080	0.887	0.326	0.143	0.120	0.750	0.332
DoRA (r=4)	5,652	0.104	0.077	0.899	0.329	0.150	0.129	0.725	0.333

7.2 Current comparison (E2)

After lowering the pretraining learning rate, the pretrained base improved markedly: zero-shot normal RMSE fell from 0.113 to 0.082 and now almost equals that of the scratch model (Table 3).

Inferences.

- **Normal events.** All five models achieve similar accuracy, with RMSE between 0.082 and 0.086 and MAE between 0.060 and 0.062. Zero-shot has the lowest mean RMSE (0.082), while LoRA and scratch both achieve 0.084. The differences are small relative to the variability across seeds, so we do

Table 3: E2 (current pipeline), raw intervals, three seeds. Trainable parameters in brackets.

		Normal				Extreme			
Model	Train. params	RMSE	MAE	PICP	MPIW	RMSE	MAE	PICP	MPIW
Zero-shot	0 (0%)	0.082 ± 0.001	0.060 ± 0.001	0.916 ± 0.015	0.261 ± 0.012	0.135 ± 0.006	0.116 ± 0.006	0.565 ± 0.048	0.272 ± 0.012
Scratch	46,460 (100%)	0.084 ± 0.004	0.062 ± 0.002	0.927 ± 0.005	0.283 ± 0.012	0.131 ± 0.004	0.110 ± 0.002	0.670 ± 0.022	0.288 ± 0.017
Full FT	46,460 (100%)	0.086 ± 0.001	0.062 ± 0.001	0.927 ± 0.018	0.288 ± 0.017	0.131 ± 0.003	0.108 ± 0.002	0.702 ± 0.026	0.295 ± 0.017
LoRA (r=4)	4,752 (10.2%)	0.084 ± 0.001	0.062 ± 0.000	0.925 ± 0.007	0.281 ± 0.009	**0.127** ± 0.001	**0.104** ± 0.000	0.721 ± 0.024	0.292 ± 0.011
DoRA (r=4)	5,652 (12.2%)	0.085 ± 0.000	0.062 ± 0.000	0.914 ± 0.001	0.270 ± 0.002	0.128 ± 0.003	0.106 ± 0.003	0.687 ± 0.022	0.284 ± 0.005

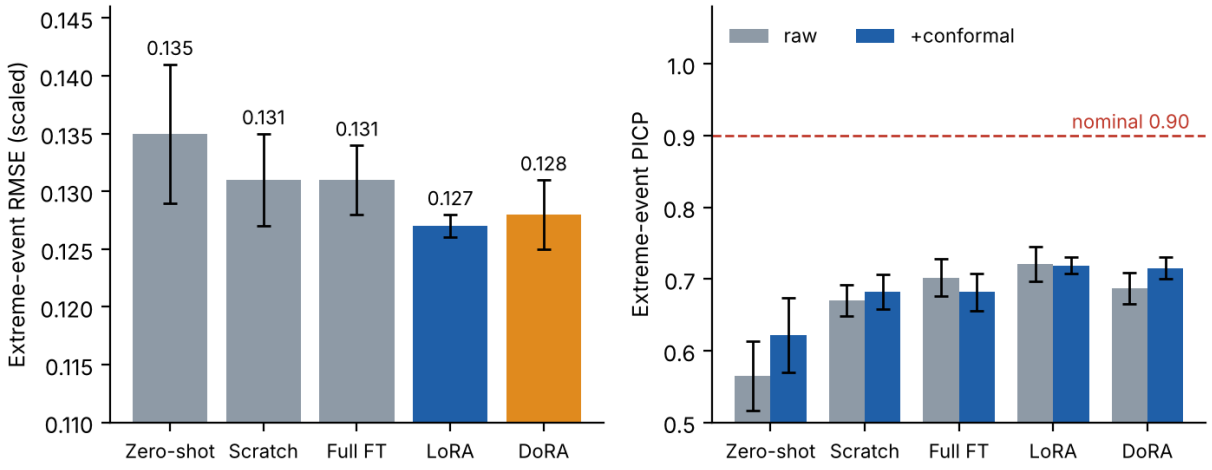


Figure 3: E2, extreme-event subset: RMSE of the median forecast (left) and interval coverage before and after conformal calibration (right). The dashed line is the nominal coverage 0.90.

not claim a meaningful accuracy advantage for any model. Raw coverage ranges from 0.914 to 0.927, close to the nominal 0.90.

- **Extreme events.** LoRA achieves the lowest mean RMSE (0.127) and MAE (0.104), followed closely by DoRA (0.128 and 0.106). Both outperform zero-shot (RMSE 0.135, MAE 0.116), scratch (0.131, 0.110) and full fine-tuning (0.131, 0.108) in mean extreme-event accuracy. LoRA's RMSE is approximately 6% lower than zero-shot's while training only 10.2% of the model parameters; DoRA trains 12.2%. This is suggestive of a benefit from parameter-efficient adaptation on extremes.

- **Pretraining.** The zero-shot model achieves the lowest normal-event RMSE (0.082), matching or improving on the adapted models, and has an extreme-event RMSE of 0.135. Scratch training and full fine-tuning improve mean extreme-event RMSE to 0.131, while LoRA and DoRA improve it further to 0.127 and 0.128. Thus, the pretrained model provides strong zero-shot performance, while adaptation—particularly LoRA and DoRA—may improve extreme-event prediction. A low-data experiment would help establish whether pretraining is especially useful when target data are limited (Section 8).

- **LoRA versus DoRA.** LoRA has slightly lower mean RMSE and MAE on extreme events (0.127 and 0.104 versus 0.128 and 0.106 for DoRA). LoRA has slightly higher normal-event coverage (0.925 versus 0.914 for DoRA). Given the small differences and variation across seeds, we do not claim a definitive advantage for either method.

7.3 Calibration and interval width

Raw extreme-event coverage is far below the nominal level (0.56–0.72). Conformal calibration on the validation year brings normal-event coverage to 0.93, but extreme-event coverage only to 0.62–0.72 (Table 4, Figure 3).

Table 4: E2 with split-conformal calibration (margin fitted on the validation year). RMSE and MAE are unchanged by calibration and are in Table 3.

	Normal		Extreme	
Model	PICP	MPIW	PICP	MPIW
Zero-shot + conformal	0.933 ± 0.002	0.279 ± 0.002	0.622 ± 0.052	0.289 ± 0.004
Scratch + conformal	0.931 ± 0.004	0.287 ± 0.001	0.682 ± 0.024	0.292 ± 0.007
Full FT + conformal	0.923 ± 0.003	0.282 ± 0.003	0.682 ± 0.026	0.289 ± 0.003
LoRA + conformal	0.924 ± 0.000	0.280 ± 0.002	**0.719** ± 0.011	0.291 ± 0.005
DoRA + conformal	0.925 ± 0.001	0.280 ± 0.000	0.715 ± 0.015	0.294 ± 0.004

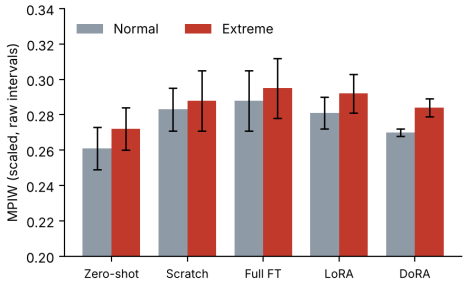


Figure 4: E2, mean interval width (raw intervals) on normal and extreme target steps.

Inferences. Split-conformal calibration brings normal-event coverage to approximately 0.923–0.933, close to the nominal 0.90, but extreme-event coverage remains lower at 0.622–0.719. LoRA achieves the highest mean extreme-event coverage (0.719), closely followed by DoRA (0.715), while zero-shot has the lowest (0.622). Calibration therefore improves coverage overall but does not achieve nominal coverage on the extreme subset.

The raw intervals remain relatively narrow on extreme events. For LoRA, MPIW increases from 0.281 on normal events to 0.292 on extreme events, a difference of approximately 4%. After conformal calibration, its MPIW increases from 0.280 to 0.291. These modest increases are insufficient to cover the extreme subset at the nominal rate.

Two factors may contribute to this result. First, the share of extreme steps rises from 5% in training to 7% in validation and 8.6% in test (Table 1), suggesting a distribution shift between the calibration and test periods. Standard split-conformal guarantees do not generally ensure coverage under such a shift or conditional on extreme outcomes. Second, coverage on the extreme subset is conditional on the target exceeding the extreme threshold; a nominal 5%–95% prediction interval does not guarantee 90% coverage within this subgroup. We therefore do not claim that the intervals adapt sufficiently to extreme volatility. Achieving higher extreme-event coverage would likely require wider intervals, so coverage and interval width must be looked at together.

7.4 Qualitative results

The Gradio interface shows, for any 2023 test window of the target locations, the history of the scaled anomaly, the true future, the median forecast, the conformally widened 90% interval and the extreme steps (orange markers). With the extreme-only option, the selected windows all contain extreme steps, and the plot shows where the truth leaves the band, which is consistent with the coverage numbers above.

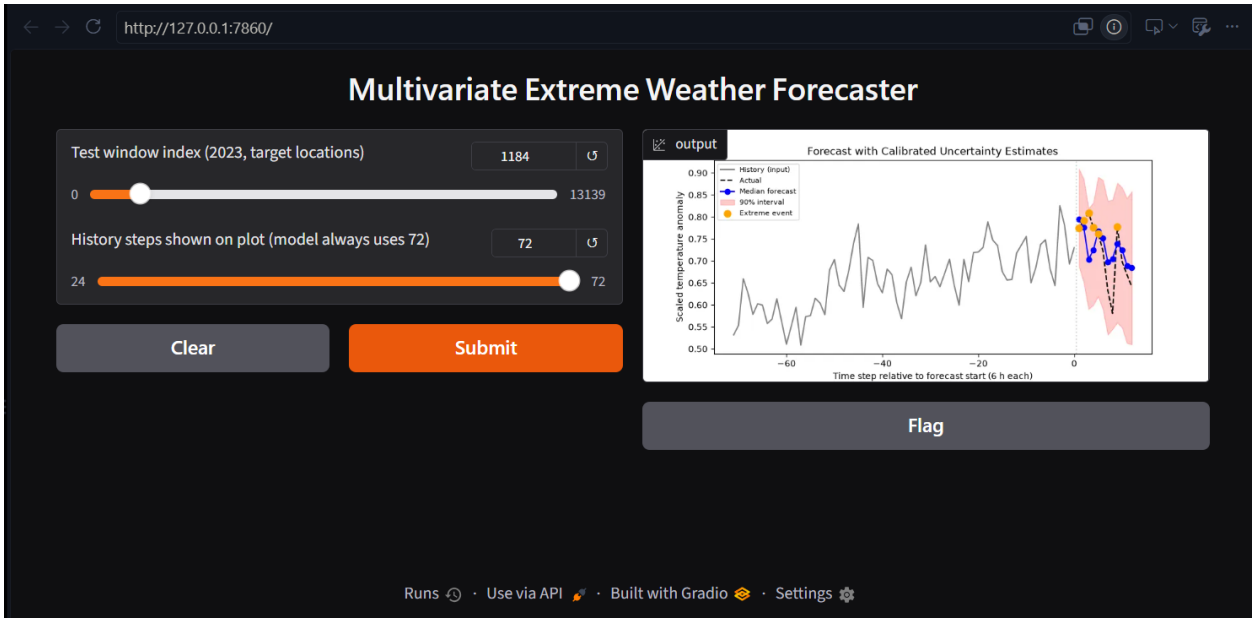


Figure 5: E2, Gradio interface on one of the selected windows, the graph shows the predicted mean value, the 90 percent confidence interval range and the extreme events (marked as orange dots)

7.5 Including cold extremes (E3)

In this experiment a target step is flagged as extreme if the temperature anomaly is above the 95th or below the 5th percentile of the training years, so cold extremes are included alongside hot ones and about 10% of the training targets are flagged. Because the normal subset now excludes both tails, it is easier, and all normal-event RMSE values are lower (0.072–0.078) than in the hot-only experiment; the two experiments therefore cannot be compared number by number. On normal events all models are accurate and the raw intervals slightly over-cover (PICP 0.935–0.954). On extreme events the five models differ little in RMSE: scratch, full fine-tuning, LoRA and DoRA all reach 0.149–0.150, within the seed standard deviation (0.002–0.005), and only zero-shot is clearly worse (0.154). In MAE all four adapted or retrained models (0.119–0.122) also beat zero-shot (0.130) by about 7–9%. The advantage of LoRA on hot extremes (RMSE 0.127 against 0.130 for scratch) therefore disappears once cold extremes are included, and LoRA, DoRA and full fine-tuning are indistinguishable; we cannot say that parameter-efficient adaptation improves extreme-event accuracy in this setting. Raw coverage on extremes is again far below the nominal 0.90 (0.565 for zero-shot, 0.675–0.708 for the other models), but the intervals widen more than before, by 7–12% on extremes against normal events (for LoRA, MPIW 0.295 on normal and 0.327 on extreme). Split-conformal calibration helps only the zero-shot model (extreme PICP 0.565 to 0.615). For the other models it *lowers* extreme coverage (for example LoRA from 0.708 to 0.673) because their raw intervals already over-cover on normal events, so the margin fitted on the validation years is negative and narrows the intervals (MPIW 0.295 to 0.280). This shows that marginal conformal calibration, which targets 90%

coverage on average over all targets, does not address under-coverage on extreme events and can make it worse, which motivates the extreme-aware calibration planned in Section 8.

Table 5: Cold extremes included, normal events (mean ± std over three seeds). RMSE and MAE are unaffected by conformal calibration.

		raw intervals				+ conformal	
Model	Train. params	RMSE	MAE	PICP	MPIW	PICP	MPIW
Zero-shot	0	0.072 ± 0.000	0.055 ± 0.000	0.935 ± 0.014	0.258 ± 0.012	0.951 ± 0.001	0.276 ± 0.002
Scratch	46,460	0.076 ± 0.003	0.057 ± 0.002	0.954 ± 0.003	0.300 ± 0.013	0.946 ± 0.002	0.288 ± 0.010
Full FT	46,460	0.078 ± 0.002	0.058 ± 0.001	0.949 ± 0.006	0.291 ± 0.008	0.940 ± 0.002	0.280 ± 0.001
LoRA (r=4)	4,752	0.077 ± 0.002	0.057 ± 0.001	0.953 ± 0.006	0.295 ± 0.010	0.941 ± 0.000	0.280 ± 0.003
DoRA (r=4)	5,652	0.077 ± 0.001	0.058 ± 0.001	0.948 ± 0.002	0.294 ± 0.003	0.938 ± 0.002	0.282 ± 0.001

Table 6: Cold extremes included, extreme events (mean ± std over three seeds). An extreme step is a temperature anomaly above the 95th or below the 5th percentile of the training years.

		raw intervals				+ conformal	
Model	Train. params	RMSE	MAE	PICP	MPIW	PICP	MPIW
Zero-shot	0	0.154 ± 0.001	0.130 ± 0.001	0.565 ± 0.027	0.288 ± 0.012	0.615 ± 0.019	0.305 ± 0.003
Scratch	46,460	0.149 ± 0.004	0.122 ± 0.003	0.675 ± 0.012	0.320 ± 0.014	0.647 ± 0.002	0.308 ± 0.011
Full FT	46,460	0.150 ± 0.005	0.119 ± 0.005	0.698 ± 0.012	0.318 ± 0.007	0.670 ± 0.030	0.307 ± 0.004
LoRA (r=4)	4,752	0.149 ± 0.002	0.120 ± 0.001	0.708 ± 0.020	0.327 ± 0.009	0.673 ± 0.007	0.312 ± 0.005
DoRA (r=4)	5,652	0.149 ± 0.002	0.120 ± 0.003	0.706 ± 0.008	0.323 ± 0.006	0.678 ± 0.006	0.311 ± 0.004

8 Plan for Novelty Assessment

The following experiments are planned for the novelty assessment.

- **Calibration for extreme events (main novelty direction).** Our intervals under-cover extreme events (raw PICP 0.57–0.72 against the nominal 0.90), and standard split-conformal calibration [10], which targets average coverage over all targets, does not fix this and can even lower extreme coverage. We plan to calibrate for extremes using only quantities known at forecast time: (a) conditional (Mondrian) conformal calibration, with a separate margin for each forecast step and bin of the predicted anomaly; (b) separate margins for the lower and upper bound, since hot and cold extremes differ; (c) calibration on the most recent months, or online adaptive conformal inference [11], to reduce the train/test shift. We will compare variants by PICP on normal and extreme events, MPIW and the interval score, and report the coverage–width trade-off.
- **Low-data adaptation.** Train scratch, full fine-tuning, LoRA and DoRA using only a fraction of the target data (for example, 10%, 25% or 50%). Compare their performance with the full-data setting to determine whether pretraining provides a greater advantage when labelled target data are limited.
- **Forecast-horizon analysis.** Evaluate performance separately at different forecast lead times within the 12-step prediction horizon. This will show whether parameter-efficient adaptation remains beneficial for longer-range forecasts and whether uncertainty increases more rapidly for extreme events.

9 Conclusion

We studied multivariate weather forecasting with calibrated uncertainty, focusing on extreme events. We built a leak-free pipeline on ERA5 data, defined extremes by a 95th-percentile peaks-over-threshold rule,

and trained a Quantile CNN-LSTM that outputs a 90% prediction interval. Following the post-review guidance, we pretrained it on 72 grid points and adapted it to 9 held-out points with LoRA and DoRA, comparing with zero-shot, scratch and full fine-tuning, and with conformal calibration. Our first implementation had leakage, a weak deseasonalisation and no pretraining, and our first re-engineered run had a weak pretrained model; fixing the pretraining learning rate was necessary to obtain a useful base model.

In the current evaluation, LoRA and DoRA match the other methods on normal events and reach the lowest extreme-event error (RMSE 0.127 and 0.128 against 0.131–0.135) with 10–12% of the trainable parameters, and the highest extreme-event coverage (0.72). The main limitation is that coverage on extreme events (0.64–0.73) stays well below the nominal 0.90, because of the distribution shift between training and test years and because extreme coverage is conditional on the outcome. We were thus, not able to achieve a high accuracy in the extreme event forecast. Future work includes testing better extreme event calibration, a low-data adaptation study, and forecast horizon analysis. (Section 8).

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